

SME21824 — MAK Algonquin Station 1 MRI

Northside · GE MRI (ge167) · Jul 20 — Jul 28 monitor data (388 captures · 46,373 readings) · analyzed 2026-07-28

COMPRESSOR

RESTARTED

recool began Jul 28 14:16Z
· off ~17.5 h

COLDHEAD

4.457 K

from 77.9 K · base
temp in 16 min

PRESSURE NOW

3.660 PSI

73% of the 5 PSI line ·
turning down

EVENT PEAK

3.728 PSI

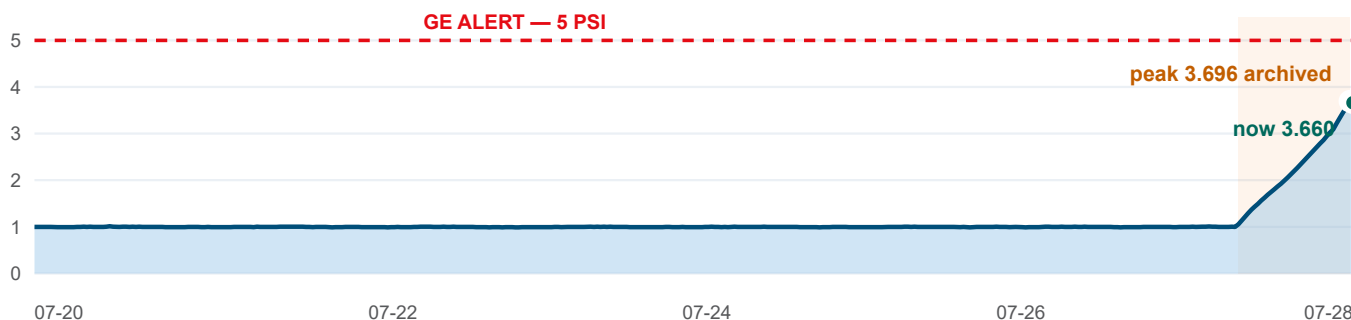
14:28Z · 1.27 PSI
under threshold

HELIUM

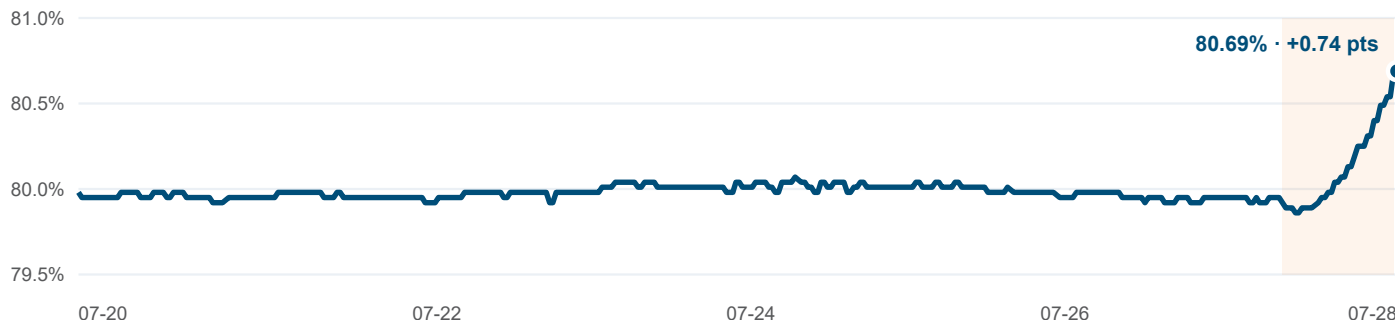
80.69%

+0.74 pts · no loss ·
no quench

HE PRESSURE — JUL 20 → JUL 28 (every archived capture, PSI · orange shade = the warm window)



HELIUM LEVEL — JUL 20 → JUL 28 (every archived capture, %)



Timeline (UTC; console clock 72 min slow — see DATA NOTES): Baseline Jul 20–27: pressure flat 0.989–1.011 PSI across 353 captures, coldhead 4.09–4.40 K, shield ~51 K, helium 79.92–79.95%. **The compressor stopped between 20:43Z and 21:13Z on Jul 27.** The magnet warmed ~17.5 h (17 h 02 m – 17 h 33 m): shield 51 → 111.6 K, coldhead 4.10 → 77.9 K, He pressure rose **strictly monotonically across all 34 warm captures**, 1.056 → 3.696 PSI (+0.155 PSI/h avg), **accelerating to +0.235 PSI/h in the final hour** — though it eased to +0.119 PSI/h mid-event, so acceleration is not uniform. **Pressure peaked 3.728 PSI at 14:28Z and never reached the 5 PSI GE threshold**, 1.272 PSI below the line. Helium rose 79.92 → 80.69% (one dip to 79.86%): **no helium lost, no quench.** **The compressor restarted between 14:15Z and 14:16Z Jul 28.** All three cold channels fell together over 16 min (76.2 → 4.9 K, 64.9 → 4.9 K, 96.3 → 4.9 K), reached base at 14:32Z and hold 4.457–4.559 K; shield 111.6 → 77.7 K. Pressure turned over at 14:29Z to 3.660 PSI. **Warming event CLOSED; pressure decay still in progress** — 3.660 PSI is 3.7× the 0.998 baseline. CDC1 (100 → 28) and HDC (30 → 4) have not recovered; EC3/EC4 read 0.

SAMPLES BEHIND THE RECOVERY

Archived log:

1 capture

(14:44Z). Run log:

19 distinct

minute readings — 17 on the ramp, 2 at base; the 14 rows after 14:31Z are 3 repeats of one value.

NOT AN ARTIFACT

Water_Flow held 0.245–0.248 and Water_Temp 47.3–51.6 °F right through the ramp — not the all-channels dropout. The cold channels fell as a smooth 16-minute monotone curve, not one step.

DATA NOTES

Console

72 min slow

vs the box (71 m 27 s – 72 m 02 s over 5 pairs; box NTP-synced). Peak

3.728 PSI

only in the minute stream. RF alternates 0/1 in healthy and warm rows alike —

not a recovery flag